

## KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3382

3 Credits: 4

Program Core

Source-grounded

# Introduction to Artificial

Students will be able to summarize the fundamental concepts of artificial intelligence and

## OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3382 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3382.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

| Item                 | Official information                                 |
|----------------------|------------------------------------------------------|
| Programme            | Diploma in Artificial Intelligence                   |
| Course code          | 3382                                                 |
| Course title         | Introduction to Artificial                           |
| Semester             | 3 Credits: 4                                         |
| Course category      | Program Core                                         |
| Periods per week     | 4 (L:4 T:0 P:0) Periods per semester: 60             |
| Periods per semester | 60                                                   |
| Credits              | 4                                                    |
| Prerequisite         | See the official syllabus PDF                        |
| Assessment           | Use the official assessment section reproduced below |

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

**STUDY METHOD****How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

**Second pass**

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

**Practical evidence**

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

**Revision pass**

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

**LEARNING OUTCOMES**

# Objectives, outcomes and mapping

## Course objectives

- Students will be able to summarize the fundamental concepts of artificial intelligence and
- solve various problems using AI techniques.

## Course outcomes

| CO  | Official outcome                                                                                      | Study level           |
|-----|-------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 15 Understanding the significance of Intelligent Agents in AI. Apply Mathematical logic for knowledge | See official syllabus |
| CO2 | 15 Applying representation of AI                                                                      | See official syllabus |
| CO3 | Practice various search techniques in AI 16 Applying                                                  | See official syllabus |
| CO4 | Describe learning and expert systems. 14 Understanding                                                | See official syllabus |

## CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3                      |
| CO2 | CO2 3                      |
| CO3 | CO3 3                      |
| CO4 | CO4 3                      |

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

| Module   | Official heading                                            | Topic entries | Hours        |
|----------|-------------------------------------------------------------|---------------|--------------|
| Module 1 | Intelligent Agents in AI.                                   | 7             | As specified |
| Module 2 | Apply Mathematical logic for knowledge representation of AI | 7             | As specified |
| Module 3 | Practice various search techniques in AI                    | 3             | As specified |
| Module 4 | Familiarize with constraint satisfaction problems in AI     | 4             | As specified |

**MODULE 1**

# Intelligent Agents in AI.

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Intelligent Agents in AI.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## ▼ Define AI, foundation and history of AI.

Define AI, foundation and history of AI. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Define AI, foundation and history of AI. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, define ai, foundation and history of ai. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What Define AI, foundation and history of AI. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Define AI, foundation and history of AI.   
 പഠിക്കേണ്ടതെന്തെന്ന് definition/meaning പറയണം.   
 പഠിക്കേണ്ടത്, steps, application പറയണം.   
 English- Define AI, foundation and history of AI.   
 Technical terms



## ▼ Discuss applications of AI.

Discuss applications of AI. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Discuss applications of AI. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, discuss applications of ai. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                       |
|---------------------|-----------------------------------------------------------------------|
| Definition/identity | What Discuss applications of AI. means in the official course context |
| Study lens          | Principle → components or stages → result → application               |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Discuss applications of AI. Discuss definition/meaning of AI. Discuss various applications, steps, application of AI in various fields. Technical terms English- Malayalam.

## ▼ Explain the Turing test.

Explain the Turing test. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the Turing test. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the turing test. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                    |
|---------------------|--------------------------------------------------------------------|
| Definition/identity | What Explain the Turing test. means in the official course context |
| Study lens          | Principle → components or stages → result → application            |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Explain the Turing test. definition/meaning . steps, application . Technical terms English- .

## ▼ Explain AI problems

Explain AI problems is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain AI problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain ai problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                               |
|---------------------|---------------------------------------------------------------|
| Definition/identity | What Explain AI problems means in the official course context |
| Study lens          | Principle → components or stages → result → application       |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Explain AI problems   
 definition/meaning . steps, application . Technical terms English-   
 .

## ▼ Explain agents and environments.

Explain agents and environments. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain agents and environments. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain agents and environments. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                            |
|---------------------|----------------------------------------------------------------------------|
| Definition/identity | What Explain agents and environments. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-എന്നു വിശദീകരിക്കുക. നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവ ഉൾപ്പെടെയുള്ളവ. Technical terms English-ൽ ഉൾപ്പെടെയുള്ളവ.



## ▼ Identify structure of agents

Identify structure of agents is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Identify structure of agents using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, identify structure of agents should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                        |
|---------------------|------------------------------------------------------------------------|
| Definition/identity | What Identify structure of agents means in the official course context |
| Study lens          | Principle → components or stages → result → application                |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ് Identify structure of agents

ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ് definition/meaning ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ്. ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ്, steps, application ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ്. Technical terms English-ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ് ഓർജനൈസ്ഡ്.

▼ **Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.**

Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, recognize types of agents. 2 understanding introduction to ai-ai concepts, foundations of ai, applications of ai -language models - information retrieval - information extraction - natural language processing - machine translation - speech recognition, turing test. ai problems: water jug problem, tower of hanoi problem, four-queens problem, travelling salesman problem intelligent agents-agents and environments, structure of agent - types of agents. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Recognize types of agents. 2  
Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. ഏതെങ്കിലും ഏതെങ്കിലും definition/meaning ഏതെങ്കിലും ഏതെങ്കിലും, steps, application ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-ൽ എഴുതുക. ഏതെങ്കിലും.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 2**

# Apply Mathematical logic for knowledge representation of AI

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Apply Mathematical logic for knowledge representation of AI
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## ▼ knowledge representation and reasoning Understanding Describe the representation and reasoning 2

knowledge representation and reasoning Understanding Describe the representation and reasoning 2 is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify knowledge representation and reasoning Understanding Describe the representation and reasoning 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, knowledge representation and reasoning understanding describe the representation and reasoning 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What knowledge representation and reasoning Understanding Describe the representation and reasoning 2 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ knowledge representation and reasoning Understanding Describe the representation and reasoning 2 □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.



## ▼ about objects, relations, events, actions, time Understanding and space

about objects, relations, events, actions, time Understanding and space is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify about objects, relations, events, actions, time Understanding and space using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, about objects, relations, events, actions, time understanding and space should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What about objects, relations, events, actions, time Understanding and space means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഓബ്ജക്റ്റുകൾ, ബന്ധങ്ങൾ, സംഭവങ്ങൾ, പ്രവൃത്തികൾ, സമയം Understanding and space ഓബ്ജക്റ്റുകളുടെ definition/meaning ഓബ്ജക്റ്റുകളുടെ. ഓബ്ജക്റ്റുകൾ ഓബ്ജക്റ്റുകൾ, steps, application ഓബ്ജക്റ്റുകൾ ഓബ്ജക്റ്റുകൾ. Technical terms English-ഓ ഓബ്ജക്റ്റുകൾ ഓബ്ജക്റ്റുകൾ.

## ▼ Demonstrate Propositional Logic

Demonstrate Propositional Logic is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate Propositional Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate propositional logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                           |
|---------------------|---------------------------------------------------------------------------|
| Definition/identity | What Demonstrate Propositional Logic means in the official course context |



## ▼ Demonstrate First Order Logic

Demonstrate First Order Logic is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate First Order Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate first order logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                         |
|---------------------|-------------------------------------------------------------------------|
| Definition/identity | What Demonstrate First Order Logic means in the official course context |
| Study lens          | Principle → components or stages → result → application                 |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നു കാണിക്കുക First Order Logic

എന്നതിന്റെ definition/meaning എഴുതുക. എന്നതിന്റെ steps, application എന്നിവ എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ Explain Soundness and Completeness,

Explain Soundness and Completeness, is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain Soundness and Completeness, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain soundness and completeness, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                               |
|---------------------|-------------------------------------------------------------------------------|
| Definition/identity | What Explain Soundness and Completeness, means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നു വിശദീകരിക്കുകയും പൂർണ്ണതയും, നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുകയും. നിർവ്വചനം നിർവ്വചിക്കുകയും, steps, application നിർവ്വചിക്കുകയും നിർവ്വചിക്കുകയും. Technical terms English-ൽ നിർവ്വചിക്കുകയും നിർവ്വചിക്കുകയും.



## ▼ Explain Forward and Backward chaining

Explain Forward and Backward chaining is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain Forward and Backward chaining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain forward and backward chaining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| Definition/identity | What Explain Forward and Backward chaining means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Explain Forward and Backward chaining  
 പരമ്പരപരമായ രീതിയിൽ definition/meaning നൽകുന്നു. പരമ്പരപരമായ  
 രീതിയിൽ, steps, application നൽകുന്നു പരമ്പരപരമായ രീതിയിൽ. Technical terms  
 English-ൽ പരമ്പരപരമായ രീതിയിൽ.

▼ **Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs.**

Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences:

Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, familiarize with classical planning 2 understanding knowledge representation and reasoning-ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. logic and inferences: propositional logic, first order logic, knowledge representation in first order logic, inference in first order logic, soundness and completeness, forward and backward chaining. classical planning - algorithms for planning state space search, planning graphs. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 3**

# Practice various search techniques in AI

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Practice various search techniques in AI
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## ▼ Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth

Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth is an official topic in Module 3 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarise types of search algorithms 2 understanding implement breadth first search and depth should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth definition/meaning . steps, application . Technical terms English- .



## ▼ 6 Applying First Search Implement Best first search, Hill Climbing, A\*

6 Applying First Search Implement Best first search, Hill Climbing, A\* is an official topic in Module 3 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Applying First Search Implement Best first search, Hill Climbing, A\* using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 applying first search implement best first search, hill climbing, a\* should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Applying First Search Implement Best first search, Hill Climbing, A* means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 6 Applying First Search Implement Best first search, Hill Climbing, A\* definition/meaning. steps, application. Technical terms English- .

## ▼ 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A\* algorithm Beam Search

6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A\* algorithm Beam Search is an official topic in Module 3 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A\* algorithm Beam Search using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 applying & beam search search algorithm terminologies, types of search algorithms uninformed search: breadth first search, depth first search informed search: heuristic search - best first search, hill climbing, a\* algorithm beam search should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A\* algorithm Beam Search   
 definition/meaning . steps, application . Technical terms English-   
 .

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 4**

# Familiarize with constraint satisfaction problems in AI

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Familiarize with constraint satisfaction problems in AI
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition

3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 & conquer approaches, beam stack search applying describe the problem decomposition should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition definition/meaning . steps, application . Technical terms English- .



## ▼ techniques - Goal Trees, AO\*, Rule Based 4 Systems Applying Compare and summarize game playing

techniques - Goal Trees, AO\*, Rule Based 4 Systems Applying Compare and summarize game playing is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify techniques - Goal Trees, AO\*, Rule Based 4 Systems Applying Compare and summarize game playing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, techniques - goal trees, ao\*, rule based 4 systems applying compare and summarize game playing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നിരുന്നിട്ടും - Goal Trees, AO\*, Rule Based 4 Systems Applying Compare and summarize game playing എന്നിവയെക്കുറിച്ചുള്ള definition/meaning നൽകുക. എന്നിവയെക്കുറിച്ചുള്ള steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

## ▼ 4 algorithms - Minimax, AlphaBeta

4 algorithms - Minimax, AlphaBeta is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 algorithms - Minimax, AlphaBeta using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 algorithms - minimax, alphabeta should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Definition/identity | What 4 algorithms - Minimax, AlphaBeta means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 4 algorithms - Minimax, AlphaBeta  
 നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക  
 നിർവ്വചനം, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms  
 English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ **Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO\*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) - Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 <https://www.tutorialspoint.com/virtualization2.0/index.htm> 2 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 3 [https://onlinecourses.swayam2.ac.in/arp19\\_ap79/preview](https://onlinecourses.swayam2.ac.in/arp19_ap79/preview)**

Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO\*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) - Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 <https://www.tutorialspoint.com/virtualization2.0/index.htm> 2

[https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 3

[https://onlinecourses.swayam2.ac.in/arp19\\_ap79/preview](https://onlinecourses.swayam2.ac.in/arp19_ap79/preview) is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Familiarize Constraint Satisfaction Problem 3  
Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO\*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 <https://www.tutorialspoint.com/virtualization2.0/index.htm> 2 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 3 [https://onlinecourses.swayam2.ac.in/arp19\\_ap79/preview](https://onlinecourses.swayam2.ac.in/arp19_ap79/preview) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, familiarize constraint satisfaction problem 3 understanding finding optimal paths -branch & bound, divide & conquer approaches problem decomposition: goal trees, ao\*, rule based system. game playing: type of games in ai - optimal decisions in games -zero sum game, game tree, minimax algorithm, alphabeta algorithm. constraint satisfaction problems(csp) – definition, example problems, constraint propagation- inference in cps, backtracking search for cps, structure of csp problems. text / reference: t/r book title / author t1 elaine rich and kevin knight. artificial intelligence, tata mcgraw hill stuart russell and peter norvig. artificial intelligence: a modern approach, 3rd t1 edition. prentice hall. r1 deepak khemani. a first course in artificial intelligence, mcgraw hill education (india) r2 stefan edelkamp and stefan schroedl. heuristic search, morgan kaufmann r3 donald a. waterman, 'a guide to expert systems', pearson education online resources: sl. no. website link 1 <https://www.tutorialspoint.com/virtualization2.0/index.htm> 2 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 3 [https://onlinecourses.swayam2.ac.in/arp19\\_ap79/preview](https://onlinecourses.swayam2.ac.in/arp19_ap79/preview) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

## Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald |

| Aspect        | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1<br><a href="https://www.tutorialspoint.com/virtualization2.0/index.htm">https://www.tutorialspoint.com/virtualization2.0/index.htm</a> 2<br><a href="https://onlinecourses.swayam2.ac.in/aic20_sp06/preview">https://onlinecourses.swayam2.ac.in/aic20_sp06/preview</a> 3<br><a href="https://onlinecourses.swayam2.ac.in/arp19_ap79/preview">https://onlinecourses.swayam2.ac.in/arp19_ap79/preview</a> means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO\*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games - Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) - Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1  
<https://www.tutorialspoint.com/virtualization2.0/index.htm> 2  
[https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 3





**RAPID REVISION****Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 2 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 3 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 4 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**OFFICIAL SELF-LEARNING****Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

**EXAMINATION PREPARATION**

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

**ACTIVE RECALL**

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

► Define or explain Define AI, foundation and history of AI.. (3 marks)

► Define or explain Discuss applications of AI.. (3 marks)

► Define or explain Explain the Turing test.. (3 marks)

► Define or explain Explain AI problems. (3 marks)

### Module 2

► Define or explain knowledge representation and reasoning  
Understanding Describe the representation and reasoning 2. (3 marks)

► Define or explain about objects, relations, events, actions, time  
Understanding and space. (3 marks)

► Define or explain Demonstrate Propositional Logic. (3 marks)

► Define or explain Demonstrate First Order Logic. (3 marks)

### Module 3

► Define or explain Summarise types of search algorithms 2

## **Understanding Implement Breadth First Search and Depth. (3 marks)**

► **Define or explain 6 Applying First Search Implement Best first search, Hill Climbing, A\*. (3 marks)**

► **Define or explain 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A\* algorithm Beam Search. (3 marks)**

## **Module 4**

► **Define or explain 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition. (3 marks)**

► **Define or explain techniques - Goal Trees, AO\*, Rule Based 4 Systems Applying Compare and summarize game playing. (3 marks)**

► **Define or explain 4 algorithms - Minimax, AlphaBeta. (3 marks)**

► **Define or explain Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO\*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) - Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson**

**Education Online Resources: Sl. No. Website Link 1****<https://www.tutorialspoint.com/virtualization2.0/index.htm> 2****[https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 3****[https://onlinecourses.swayam2.ac.in/arp19\\_ap79/preview](https://onlinecourses.swayam2.ac.in/arp19_ap79/preview). (3 marks)**

**PRACTICE ONLY**

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Define AI, foundation and history of AI..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **knowledge representation and reasoning Understanding Describe the representation and reasoning 2.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth.**

### Module 4

1-mark definition: State the meaning of **3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.



**REFERENCE VOCABULARY**

# Glossary

**Study glossary**

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

**SOURCE-SUPPORTED REFERENCES**

## References and web resources

**References listed in the official PDF**

References are included in the official source PDF.

**Web resources listed in the official PDF**

- Sl. No. Website Link
- <https://www.tutorialspoint.com/virtualization2.0/index.htm>
- [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview)
- [https://onlinecourses.swayam2.ac.in/arp19\\_ap79/preview](https://onlinecourses.swayam2.ac.in/arp19_ap79/preview)

**IN-PAGE SEARCH****Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 3382. Verify institutional updates against the current official curriculum.